hash-turbo User Manual

> Version 1.0.1

Cross-platform file hash management tool with CLI and PySide6 QML GUI. Generates, stores, verifies, and sanitizes file hashes with support for many algorithms.

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Quick Start

```
# Hash a file
hash-turbo hash report.pdf

# Hash a directory recursively, write to a file
hash-turbo hash src/ -r -o checksums.sha256

# Verify files against a hash file
hash-turbo verify checksums.sha256
```

```
# Launch the GUI
hash-turbo gui
# Or just:
hash-turbo
```

Supported Algorithms

hash-turbo supports all major hash algorithms via Python's `hashlib`:

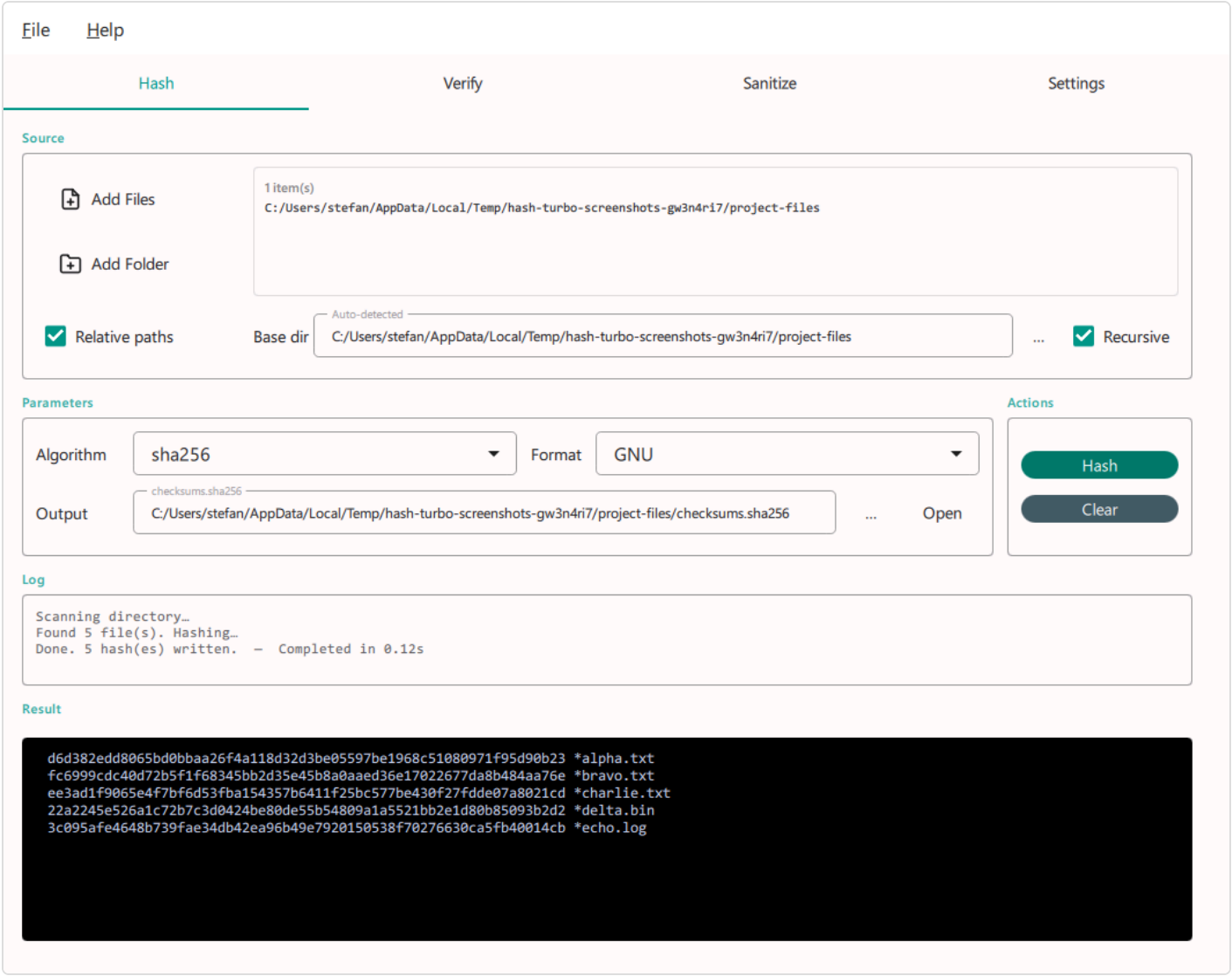
Algorithm	CLI flag	Notes
MD5	`md5`	Fast, not collision-resistant
SHA-1	`sha1`	Legacy, not recommended for security
SHA-224	`sha224`	Truncated SHA-256
SHA-256	`sha256`	**Default** — widely used, recommended
SHA-384	`sha384`	Truncated SHA-512
SHA-512	`sha512`	Stronger variant of SHA-2
SHA3-256	`sha3-256`	SHA-3 family
SHA3-512	`sha3-512`	SHA-3 family
BLAKE2b	`blake2b`	Fast, modern
BLAKE2s	`blake2s`	Fast, optimized for 32-bit

Run `hash-turbo algorithms` to list all available algorithms at runtime.

GUI

The GUI launches automatically when you run `hash-turbo` without a subcommand, or explicitly with `hash-turbo gui`. It uses a Material Design theme with four tabs: **Hash**, **Verify**, **Sanitize**, and **Settings**.

Hash View



The Hash view generates file hashes and writes them to a hash file.

Source

Add files to hash using any of these methods:

- **Add Files** button — opens a file picker for individual files
- **Add Folder** button — opens a folder picker to add an entire directory
- **Drag and drop** — drop files or folders onto the drop zone

The drop zone displays the number of pending items and lists their paths. When you add a folder, the **Base dir** field auto-fills with the folder path.

Options

- **Relative paths** checkbox — when checked, paths in the output hash file are relative to the base directory (default: checked)

- **Base dir** field — the root directory for computing relative paths; auto-detected from the first added folder
- **Recursive** checkbox — scan subdirectories (default: checked)

Parameters

- **Algorithm** — select the hash algorithm from the dropdown (default: SHA-256, configurable in Settings)
- **Format** — choose between GNU and BSD output format (default: GNU, configurable in Settings)
- **Output** — path to the output hash file; auto-filled as `<base_dir>/checksums.<algorithm>` when a folder is added. Click **...** to browse, or **Open** to open the generated file after hashing.

Actions

- **Hash** — start hashing all pending files. The button changes to **Cancel** while hashing is in progress.
- **Clear** — remove all pending files and reset the form.

Log

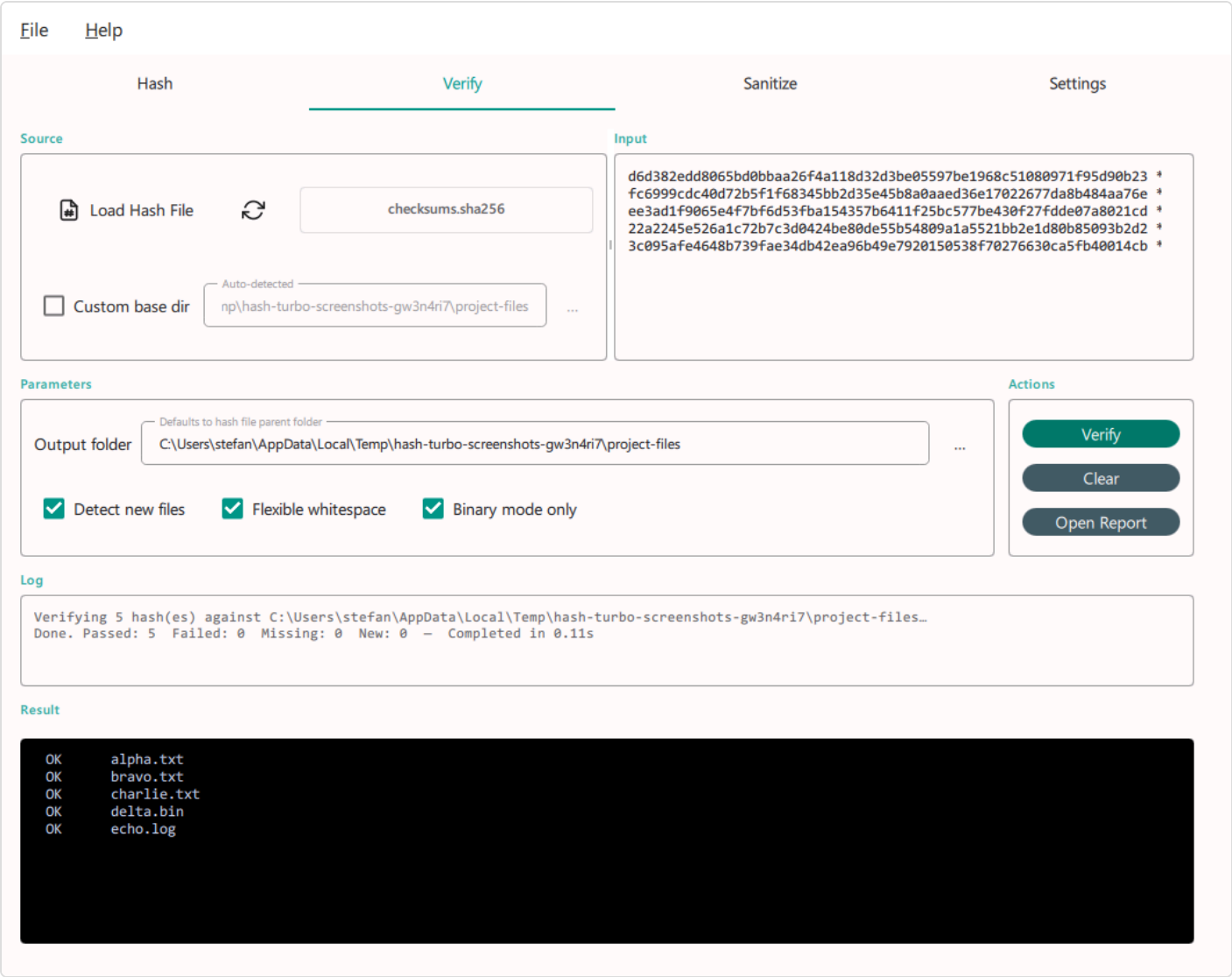
Displays real-time progress during hashing:

- A progress bar with percentage and current file name
- Status messages: scanning progress, file count, completion summary
- Final line includes elapsed time (e.g. `Done. 5 hash(es) written. — Completed in 1.23s`)

Result

A monospace terminal panel (black background) showing the generated hash entries as they are computed. Displays the last 200 lines in a rolling view. Output follows the selected format (GNU or BSD).

Verify View



The Verify view checks file integrity by comparing computed hashes against expected values from a hash file.

Source

Load a hash file using any of these methods:

- **Load Hash File** button — opens a file picker (filters: `.sha256`, `.md5`, `.sha1`, `.sha512`)
- **Drag and drop** — drop a hash file onto the drop zone
- **Reload** button (appears after loading) — re-reads the file from disk

The loaded file name is displayed in the drop zone and can be clicked to open it in the default application.

Options

- **Custom base dir** checkbox — override the auto-detected base directory for resolving file paths
- Base directory field — where to look for the files listed in the hash file (defaults to the hash file's parent directory)

Input

A read-only monospace text area showing the content of the loaded hash file. For large files (200+ entries), only a preview is shown with an entry count indicator.

Parameters

- **Output folder** — directory for verification report output (defaults to the hash file's parent)
- **Detect new files** — scan for files not listed in the hash file (default: checked)
- **Flexible whitespace** — accept tabs and multiple spaces in GNU format entries (default: checked)
- **Binary mode only** — always hash in binary mode, ignoring the text/binary indicator (default: checked)

Actions

- **Verify** — start verification. Changes to **Cancel** while running.
- **Clear** — reset all fields and results.
- **Open Report** — open the generated verification report file (enabled after verification completes).

Log

Shows verification progress with:

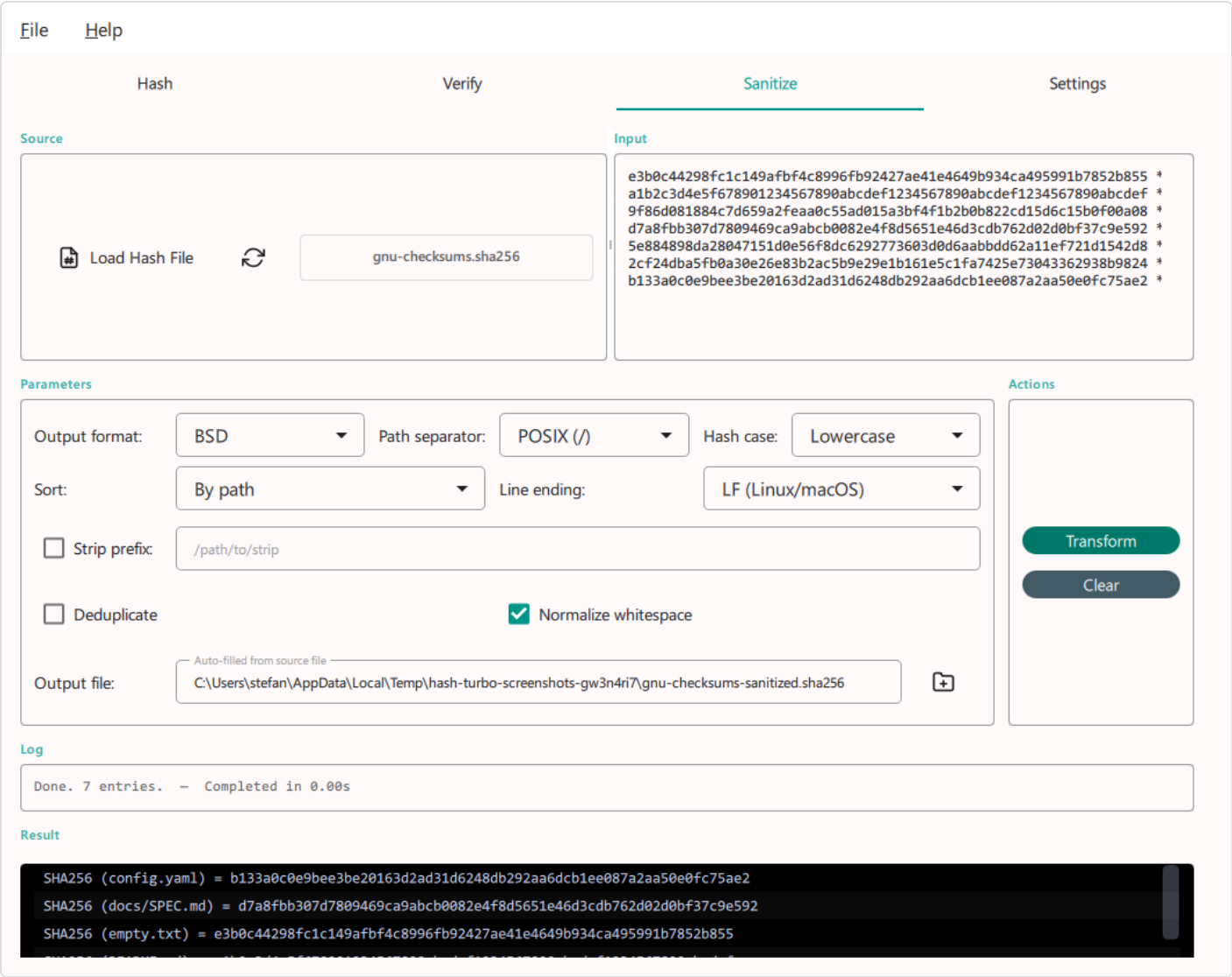
- A progress bar with percentage and current file name
- Status messages for each file being verified
- Summary line: `Passed: N Failed: N Missing: N New: N — Completed in X.XXs`

Result

A monospace terminal panel showing per-file results with status indicators:

- `OK` — hash matches
- `FAIL` — hash mismatch (shows expected vs. computed)
- `MISSING` — file not found on disk
- `NEW` — file exists but is not listed in the hash file (when "Detect new files" is enabled)

Sanitize View



The Sanitize view transforms hash manifest files — converting between formats, normalizing paths, and cleaning up entries.

Source

Load a hash file using any of these methods:

- **Load Hash File** button — opens a file picker
- **Drag and drop** — drop a hash file onto the drop zone
- **Reload** button (appears after loading) — re-reads the file from disk

Input

A read-only monospace text area showing the loaded hash file content. For large files (200+ entries), only a preview is shown.

Parameters

Parameter	Options	Description
Output format	GNU, BSD	Target format for the transformed output
Path separator	Keep original, POSIX (`/`), Windows (`\`)	Normalize path separators in file paths
Hash case	Keep original, Lowercase, Uppercase	Normalize hex digest casing
Sort	None, By path, By hash, Filesystem	Sort entries by the selected key
Line ending	System, LF, CRLF, CR	Control line ending style in output
Strip prefix checkbox + field	Text	Remove a leading path prefix from all entries
Deduplicate checkbox	On/Off	Remove duplicate entries by path
Normalize whitespace checkbox	On/Off	Fix irregular whitespace in GNU format (default: checked)

Output file

- Auto-filled as `<stem>—sanitized.<ext>` when a file is loaded
- Click **...** to browse for a custom output path
- Click the file icon to open the output file after transformation

Actions

- **Transform** — apply all selected transformations. Changes to **Cancel** while running.
- **Clear** — reset all fields and results.

Log

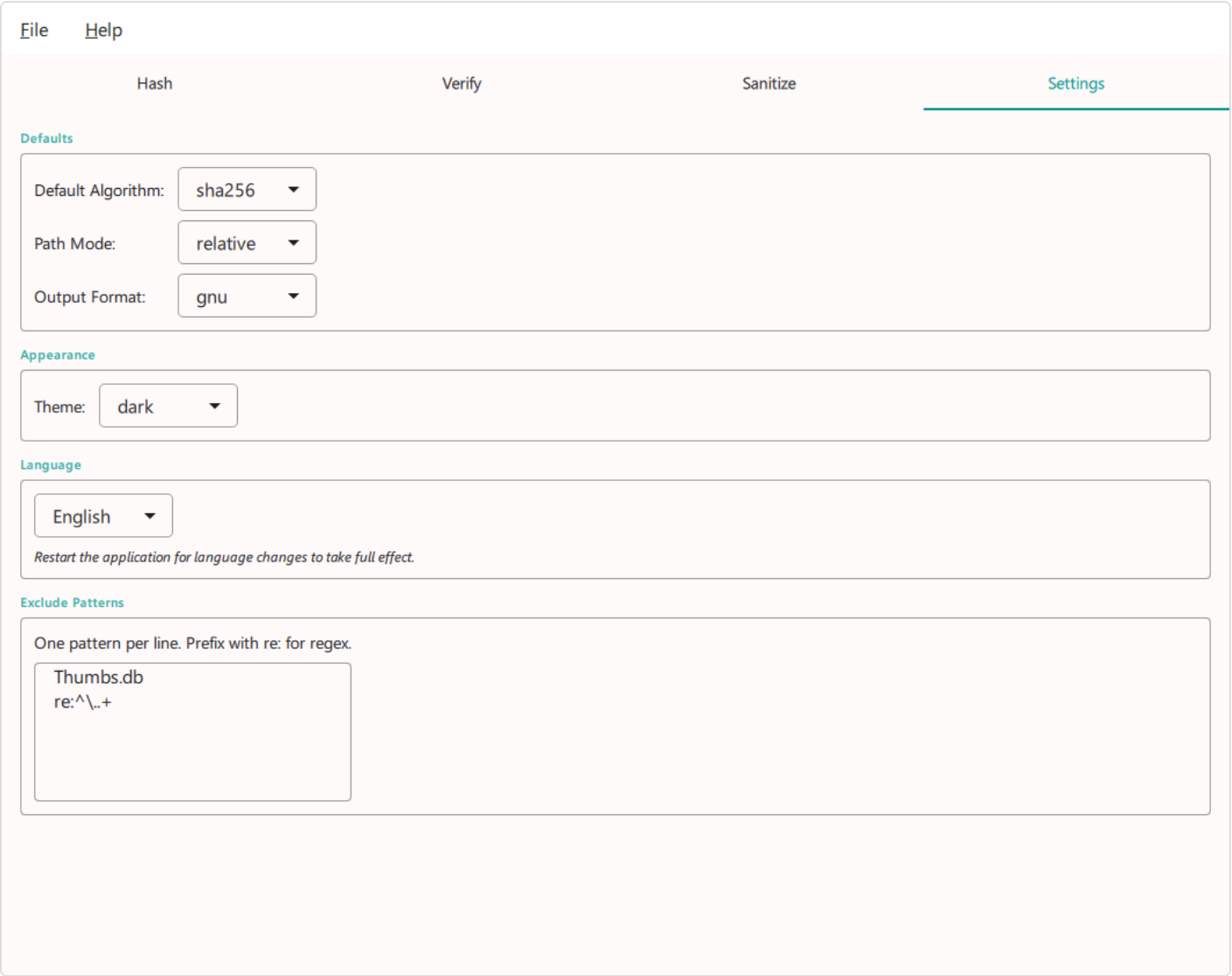
Displays status messages after each transformation:

- Completion summary with entry count and elapsed time (e.g. `Done. 7 entries. — Completed in 0.05s`)
- Error messages if the input cannot be parsed

Result

A monospace terminal panel (black background) showing the transformed hash entries in a list view. Rows have alternating backgrounds for readability. Output is auto-saved to the configured output file.

Settings



The Settings view configures application defaults. All settings are persisted across sessions.

Defaults

- **Default Algorithm** — the hash algorithm pre-selected in the Hash view (default: SHA-256)
- **Path Mode** — relative or absolute paths in hash output (default: relative)
- **Output Format** — GNU, BSD, or JSON (default: GNU)

Appearance

- **Theme** — System (follows OS), Light, or Dark

Language

- **Language selector** — English, Deutsch, Français, Italiano, Rumantsch
- Changes require an application restart to take full effect
- Translations powered by gettext (`.po` / `.mo` files)

Exclude Patterns

Define filename patterns to skip during hashing and verification:

- One pattern per line
- Supports fnmatch glob syntax (e.g., `Thumbs.db`, `*.tmp`)
- Prefix with `re:` for regex patterns (e.g., `re:^\..+` to skip hidden files)
- Defaults: `Thumbs.db`, `re:^\..+`

CLI

The CLI provides the same core features as the GUI, optimized for scripting and pipelines.

hash — Generate Hashes

```
hash-turbo hash <paths ... > [options]
```

Generate hashes for one or more files or directories.

Arguments

Argument	Description
`<paths ... >`	One or more file or directory paths (required)

Options

Option	Description	Default
`-a`, `--algorithm <algo>`	Hash algorithm	`sha256`
`-r`, `--recursive`	Recurse into directories	Off
`-g`, `--glob <pattern>`	Filter files by glob pattern	None
`--exclude <pattern>`	Exclude files matching pattern	None
`-o`, `--output <file>`	Write hashes to a file	stdout
`--format {gnu,bsd,json}`	Output format	`gnu`
`--path-mode {relative,absolute}`	Path style in output	`relative`
`--base-dir <dir>`	Base directory for relative paths	hash file parent
`-j`, `--jobs <n>`	Number of parallel workers	CPU count

Option	Description	Default
`-v` , `--verbose`	Verbose output	Off
`-q` , `--quiet`	Quiet output	Off

Examples

```
# Hash a single file (friendly display)
hash-turbo hash report.pdf
# → SHA-256: 3a7bd3e2 ... af91c  report.pdf

# Hash a directory, write GNU format
hash-turbo hash src/ -r -o checksums.sha256

# BSD format with MD5
hash-turbo hash *.zip -a md5 --format bsd

# JSON output
hash-turbo hash build/ -r --format json -o hashes.json

# Parallel hashing with 8 workers
hash-turbo hash data/ -r -j 8 -o checksums.sha256

# Only Python files, excluding tests
hash-turbo hash src/ -r -g "*.py" --exclude "test_*"
```

---

verify — Verify Hashes

```
hash-turbo verify <hashfile> [options]
hash-turbo verify <file> --expect <hash> [-a <algo>]
```

Verify files against a hash file, or verify a single file against an expected hash.

Options

Option	Description	Default
`--expect <hash>`	Verify a single file against this hash	None
`--strict`	Fail on missing files	Off
`-a` , `--algorithm <algo>`	Algorithm for `--expect` mode	`sha256`

Option	Description	Default
`--flexible-whitespace` / `--no-flexible-whitespace`	Accept irregular whitespace in GNU format	On
`--binary-only` / `--no-binary-only`	Always hash in binary mode	On
`-v` , `--verbose`	Show expected/computed hashes on failure	Off
`-q` , `--quiet`	Only show failures	Off

Exit Codes

Code	Meaning
`0`	All files verified OK
`1`	One or more mismatches or missing files (with `--strict`)
`2`	Usage error (no hash file, no entries found)

Examples

```
# Verify all files in a hash file
hash-turbo verify checksums.sha256

# Strict mode — fail on missing files
hash-turbo verify checksums.sha256 --strict

# Quiet mode — only show failures
hash-turbo verify checksums.sha256 -q

# Inline verification of a single file
hash-turbo verify download.zip --expect 3a7bd3e2360a3d29eea436fcfb7e44c735d117c42d1c1835420b6b9942

# Verbose — show expected vs computed on failure
hash-turbo verify checksums.sha256 -v
```

---

sanitize — Transform Hash Files

```
hash-turbo sanitize <hashfile> [options]
```

Transform a hash manifest file: convert formats, normalize paths, deduplicate, and sort.

Options

Option	Description	Default
`--format {gnu,bsd}`	Output format	Auto-detect from input
`--separator {keep,posix,windows}`	Path separator style	`keep`
`--strip-prefix <path>`	Remove leading path prefix	None
`--hash-case {keep,lower,upper}`	Hex digest casing	`keep`
`--sort {none,path,hash,filesystem}`	Sort entries	`none`
`--deduplicate`	Remove duplicate entries by path	Off
`--line-ending {system,lf,crlf,cr}`	Line ending style	`system`
`--normalize-whitespace` / `--no-normalize-whitespace`	Fix irregular whitespace	On
`-o`, `--output <file>`	Write result to file	stdout

Examples

```
# Convert GNU to BSD format
hash-turbo sanitize checksums.sha256 --format bsd

# Normalize to POSIX paths with lowercase hashes
hash-turbo sanitize checksums.sha256 --separator posix --hash-case lower

# Strip a path prefix to make entries relative
hash-turbo sanitize checksums.sha256 --strip-prefix C:/projects/myapp

# Deduplicate and sort, write to a new file
hash-turbo sanitize checksums.sha256 --deduplicate --sort path -o cleaned.sha256

# Full cleanup pipeline
hash-turbo sanitize checksums.sha256 \
    --format bsd \
    --separator posix \
    --hash-case lower \
    --sort path \
    --deduplicate \
    --line-ending lf \
    -o normalized.sha256
```

algorithms — List Algorithms

```
hash-turbo algorithms
```

Lists all available hash algorithms at runtime. Includes all algorithms supported by the platform's `hashlib`.

Hash File Formats

hash-turbo supports two standard hash file formats and JSON:

GNU Format

The default format, compatible with `sha256sum`, `md5sum`, etc.

```
3a7bd3e2360a3d29eea436fcfb7e44c735d117c42d1c1835420b6b9942dd4f1b *src/main.py
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 *README.md
```

- `<hex_digest> *<path>` (binary mode) or `<hex_digest> <path>` (text mode)
- The `*` prefix indicates binary mode

BSD Format

```
SHA256 (src/main.py) = 3a7bd3e2360a3d29eea436fcfb7e44c735d117c42d1c1835420b6b9942dd4f1b
SHA256 (README.md) = e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
```

- `<ALGORITHM> (<path>) = <hex_digest>`

JSON Format

Available for CLI output only (`--format json`):

```
[
  {
    "path": "src/main.py",
    "algorithm": "sha256",
    "hash": "3a7bd3e2 ... "
  }
]
```

Auto-Detection

hash-turbo automatically detects the format when reading hash files. Both GNU and BSD formats are recognized regardless of file extension.

Distribution

hash-turbo is distributed as two standalone executables built with PyInstaller:

Binary	Console	Purpose
`hash-turbo`	Yes	CLI usage — full terminal stdin/stdout/stderr support
`hash-turbo-gui`	No	GUI — launches without a console window

Both binaries bundle the complete Python runtime and all dependencies. No installation required — just download and run.